

Exploring the Application of Multi-Sensory Device Interaction in Traditional Chinese Medicine Popular Science Education.

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ABSTRACT

The origins of traditional Chinese medicine (TCM) span thousands of years, rooted in extensive medical and life experiences. Central to TCM is herbal medicine, integrating natural science principles. However, accessibility to fresh herbs is limited due to growth conditions, regulations, and transportation costs. To address this, we propose the "TCM Herbal Education" system, merging 3D scanning, interactive VR, and olfactory devices to educate on Chinese herbal science multidisciplinary. It offers insights into cultivation, pest control, prescriptions, and herbal aromas. By virtualizing traditional education, users interact with herbs via VR, enhancing understanding through sensory experiences. Exploratory research evaluates and refines the system for improved cross-reality experiences.

CCS CONCEPTS

• **Human-centered computing** → **Interaction paradigms**; Virtual Reality.

KEYWORDS

Multi-Domain Device Interaction, Virtual Reality, Traditional Chinese Medicine Education

ACM Reference Format:

Chin-Chun Chen, Tung-Hua Yang, Cheng-Ta Yang, Chi-Yu Lin, Ping-Hsuan Han, Si-Han Chen, Yi-Ru Yang, and Ching-Hui Chung. 2024. Exploring the Application of Multi-Sensory Device Interaction in Traditional Chinese Medicine Popular Science Education.. In *Special Interest Group on Computer Graphics and Interactive Techniques Conference Posters (SIGGRAPH Posters)*.

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SIGGRAPH Posters '24, July 27–August 01, 2024, Denver, CO, USA

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ACM ISBN 979-8-4007-0516-8/24/07

<https://doi.org/10.1145/3641234.3671027>

'24), July 27–August 01, 2024, Denver, CO, USA. ACM, New York, NY, USA, 2 pages. <https://doi.org/10.1145/3641234.3671027>

1 INTRODUCTION

Research on the use of VR and AR in the medical field primarily focuses on enhancing participants' understanding of illnesses and familiarizing them with medical procedures through visual means. The goal is to strengthen participants' comprehension of medical training processes and disease understanding. For example, [Bush 2008] explored the efficacy of VR exposure therapy in treating inner anxieties and fears such as acrophobia, claustrophobia, and social anxiety disorder. [Washington et al. 2016] investigated the use of AR to assist children with autism spectrum disorder in interpreting emotions of others, aiming to help these children develop healthy social relationships comparable to those of neurotypical individuals. In contemporary research and creative endeavors, there is a growing exploration of multi-sensory applications, integrating VR and AR to generate new developments. [Casalegno 2020] created the "A Virtual Feast" project: the cooking device Aerobanquets RMX, which employs VR to produce experiential processes involving visual, olfactory, and gustatory sensations. Through this project, participants undergo an experience that prompts them to reevaluate their previous visually-oriented judgments of food, thereby fostering a reevaluation of their own sensory perceptions. [Ortiz-Catalan et al. 2016] proposed using virtual environments to re-establish tactile sensations in patients with amputations, activating the brain regions responsible for limb control to achieve therapeutic effects. Addressing the demand for multisensory integration in the realm of TCM science and education, we propose the "TCM Herbal Education" system. This system utilizes digital technology to recreate the growth environment of Chinese herbs and simulate clinical experiences in a VR setting (see Figure 1c). Our aim is to bridge the gap in previous TCM education, which relied solely on images to understand the unprocessed state of Chinese herbs. By enhancing comprehension of the native ecological environments of Chinese



Figure 1: (a) Simulating the growth environment of Chinese medicinal herbs, presented in VR to depict various states of cultivation ecology and pest control. (b) High-resolution 3D scanning of raw and processed medicinal herbs. (c) Users experience the aroma of herbal essential oils using aroma spray devices. (d) Participants engage in the experience of traditional Chinese medicine symptom analysis and selection of herbs.

herbs, we aim to explore the potential for interactive engagement within the system.

2 DESIGN AND IMPLEMENTATION

The "Traditional Chinese Medicine (TCM) Herbal Education" system offers immersive virtual experiences in herbal cultivation and TCM treatment. Utilizing VR and scent technologies, users explore these environments, gaining insights into unprocessed Chinese herbs. Simulated mint garden climates and pest control methods provide educational context. Collaborating with TCM practitioners ensures accurate depiction of patient characteristics, symptom analysis, and herbal properties, meeting pedagogical standards. Users engage in interactive symptom consultations with digital patients, guided by TCM practitioners to select appropriate herbs. Olfactory cues enhance the experience, while visual effects generated using Unity are projected in real-time. Cloud-based data recording tracks experience metrics like duration, error frequency, and progression, aiding future enhancements.

3 EXPLORATORY AUDIENCE STUDY

To explore user interaction, we recruited 22 participants (12 males, 10 females, aged 32–56), including 6 with prior exposure to digital medical materials and 3 VR users. Conducted at DailyHealth Chinese Medicine Pharmacy, participants performed 5 medication retrieval tests using our system. A live test included students, teachers, TCM vendors, and buyers. After explaining the system, participants used essential oil diffusers and VR headsets, received voice instructions, and completed tasks with a time limit. Post-experience, they completed a Likert scale questionnaire and interviews.

4 DISCUSSION AND FUTURE WORK

During the (TCM Herbal Education) program, participants valued their prior experiences with Chinese herbal medicine, relying on herb color and taste for accuracy. They offered intuitive feedback

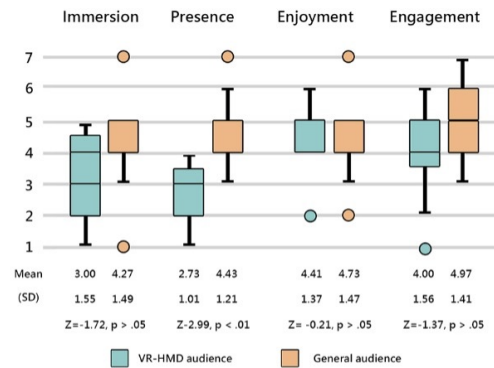


Figure 2: The statistical analysis result of the questionnaire.

on digital patients' symptoms and prescriptions. While engaged in testing challenges, they actively provided reviews and suggestions for operational errors (Figure 1d). Yet, some struggled to link original herb appearances with processed forms, diverting attention and limiting past experience application. Although most found the experience enlightening (Figure 1b), some felt the unprocessed herb state unnecessary. This study introduces a Chinese medicine diagnosis and treatment system with various devices and modes, offering immersive experiences in authentic herbal environments and treatment methods. We believe users interested in virtual interactions would embrace VR headsets for detailed herbal knowledge. However, Figure 2. transitioning realities decreased viewer identification with medical behaviors, and wearing VR devices in medical testing reduced user willingness. AR is considered to enhance engagement and allay concerns. Future plans involve aligning with the "Traditional Chinese Medicine (TCM) Herbal Education" system to promote deeper understanding via convenient mobile devices.

ACKNOWLEDGMENTS

This research was partially supported by the National Science and Technology Council of Taiwan under the grant NSTC 112-2622-H-038 -001

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